

Cheat Sheet for Interaction Concepts

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Abstract

1 Preliminaries

Let y denote the event of success (e.g. number of downloads is larger than a given threshold) and let $\bar{y}$ denote its complementary event, indicating a failure. Moreover, let r denote the event that a feature has a certain value (e.g. an app is free) and $\bar{r}$ denote its complementary event (the game is not free). Similarly, c and $\bar{c}$ denote those for another feature.

We use the notation n for the overall number of data, and its subscripts to denote the number of data for which the above conditions hold. For example, n_y and $n_{\bar{y}} = n - n_y$ denote the number of successful and not successful apps, respectively. With the counts, the probability estimates are well defined. For example, the probability of success:

$$p_y = \frac{n_y}{n} = \frac{n_y}{n_y + n_{\bar{y}}}.$$

This quantity can also be expressed in terms of **odds**, obtained by the odds transform of the probability:

$$o_y = \text{odds}(p_y) = \frac{p_y}{1 - p_y} = \frac{p_y}{p_{\bar{y}}} = \frac{n_y}{n_{\bar{y}}}$$

Both probabilities and odds can, in turn, be expressed on the **logarithmic scales**. The **logarithmic counterpart of probability is the information content**:

$$i_y = I(p_y) = -\log p_y = \log n - \log n_y$$

and that of the odds, the **log-odds** is obtained by the **logit transform of the probability**:

$$l_y = \text{logit}(p_y) = \log \frac{p_y}{1 - p_y} = \log n_y - \log n_{\bar{y}} = i_{\bar{y}} - i_y$$

For these quantities (except information content), we have the following analogous identities:

$$p_y = 1 - p_{\bar{y}} \tag{1}$$

$$o_y = \frac{1}{o_{\bar{y}}} \tag{2}$$

$$l_y = -l_{\bar{y}}$$

Likewise, n_{yc} and $n_{y\bar{c}} = n_y - n_{yc}$ denote the number of successful apps with the first feature value and those without it, respectively. These counts can be conveniently considered in a 2x2 table:

$$\begin{pmatrix} n_{yc} & n_{y\bar{c}} \\ n_{\bar{y}c} & n_{\bar{y}\bar{c}} \end{pmatrix}$$

whose row sums are $n_y = n_{yc} + n_{y\bar{c}}$ and $n_{\bar{y}} = n_{\bar{y}c} + n_{\bar{y}\bar{c}}$ and column sums $n_c = n_{yc} + n_{\bar{y}c}$ and $n_{\bar{c}} = n_{y\bar{c}} + n_{\bar{y}\bar{c}}$.

r and c are for row and column, respectively, but other symbols can also be used.

These should be put next to the table for convenience

From this table of counts, we get estimates of the **joint probabilities**, such as

$$p_{yc} = \frac{n_{yc}}{n} = \frac{n_{yc}}{n_{yc} + n_{y\bar{c}} + n_{\bar{y}c} + n_{\bar{y}\bar{c}}}$$

that can, in turn, be collected into a 2x2 table of joint probability estimates:

$$\begin{pmatrix} p_{yc} & p_{y\bar{c}} \\ p_{\bar{y}c} & p_{\bar{y}\bar{c}} \end{pmatrix}$$

whose **row sums equal the probabilities of success and failure** and the **column sums** to the probabilities of feature c being present or absent.

From the counts, we can also estimate **conditional probabilities**, such as the probability of y given that c takes place:

$$p_{y|c} = \frac{n_{yc}}{n_c} = \frac{n_{yc}}{n_{yc} + n_{\bar{y}c}}.$$

Again, these can be conveniently considered in a 2x2 table of conditional probabilities:

$$\begin{pmatrix} p_{y|c} & p_{y|\bar{c}} \\ p_{\bar{y}|c} & p_{\bar{y}|\bar{c}} \end{pmatrix}$$

whose **column sums are equal to 1 due to the identities (1) and (2)**. Hence, this table can be expressed in more condensed form with just the two conditional probabilities $p_{y|c}$ and $p_{y|\bar{c}}$ or with the corresponding conditional odds:

$$o_{y|c} = \frac{p_{yc}}{1 - p_{yc}} \quad o_{y|\bar{c}} = \frac{p_{y\bar{c}}}{1 - p_{y\bar{c}}}$$

In the context of confusion matrices, $p_{y|c}$ is known as the **positive predictive value** or the **precision** (also **risk** in epidemiological contexts), $p_{\bar{y}|c}$ as the false omission rate, $p_{y|\bar{c}}$ as the false discovery rate and $p_{\bar{y}|\bar{c}}$ as the negative predictive value.

As summary statistics, we may consider what is called the **difference in proportions**:

$$p_{y|c} - p_{y|\bar{c}}$$

or **ratio of proportions**:

$$\frac{p_{y|c}}{p_{y|\bar{c}}}$$

Analogous to the ratio of proportions, we have the ratio of odds:

$$\theta_{yc} = \frac{o_{y|c}}{o_{y|\bar{c}}} = \frac{p_{y|c}p_{\bar{y}|\bar{c}}}{p_{\bar{y}|c}p_{y|\bar{c}}} = \frac{p_{yc}p_{\bar{y}\bar{c}}}{p_{\bar{y}c}p_{y\bar{c}}} = \frac{n_{yc}n_{\bar{y}\bar{c}}}{n_{\bar{y}c}n_{y\bar{c}}}$$

For the odds-ratio, we have the following identities:

$$\theta_{yc} = 1/\theta_{\bar{y}c} = 1/\theta_{y\bar{c}} = \theta_{\bar{y}\bar{c}}$$

If $p_{yc} \approx p_y p_c$ If $p_{y|c} \approx p_{y|\bar{c}} \approx p_y$, then y can be considered to be independent of c .

With three subscripts like n_{yrc} , we denote the number of data split according to another feature r . We thus have altogether the following eight types of data:

$$\begin{pmatrix} n_{yrc} & n_{y\bar{r}c} \\ n_{\bar{y}rc} & n_{\bar{y}\bar{r}c} \end{pmatrix} \quad \begin{pmatrix} n_{\bar{y}rc} & n_{\bar{y}\bar{r}c} \\ n_{yrc} & n_{y\bar{r}c} \end{pmatrix}$$

We may consider the association between y and c within the subset determined by r :

$$\begin{pmatrix} p_{yc|r} & p_{y\bar{c}|r} \\ p_{\bar{y}c|r} & p_{\bar{y}\bar{c}|r} \end{pmatrix}$$

for which we can compute e.g. odds-ratios conditional to r :

$$\theta_{yc|r} = \frac{p_{yc|r}p_{\bar{y}\bar{c}|r}}{p_{\bar{y}c|r}p_{y\bar{c}|r}} = \frac{n_{ycr}n_{\bar{y}\bar{c}r}}{n_{\bar{y}cr}n_{y\bar{c}r}} \quad (3)$$

and for the complementary subset $\bar{r}$ analogously. If we have $\theta_{yc|r} = \theta_{yc|\bar{r}} = 1$, then y is conditionally independent of c given r .

Similarly, we can consider the association between y and r within the subsets determined by c via the analogous conditional odds-ratios $\theta_{yr|c}$ and $\theta_{yr|\bar{c}}$. If they are both 1, then y is conditionally independent of r given c . Note that we could also consider the association between r and c given y in a similar way, possibly having a third type of conditional independence.

If $\theta_{yc|r} = \theta_{yc|\bar{r}}$, that is, the values of conditional odds-ratios are the same but not necessarily 1, then we say that the associations are homogeneous. Consider the "ratio of odds-ratios":

$$\frac{\theta_{yc|r}}{\theta_{yc|\bar{r}}} = \frac{p_{yc|r}p_{\bar{y}\bar{c}|r}}{p_{\bar{y}c|r}p_{y\bar{c}|r}} = \frac{n_{ycr}n_{\bar{y}\bar{c}r}}{n_{\bar{y}cr}n_{y\bar{c}r}} \quad (4)$$

that in case of homogeneous associations is 1, implying that we must also have $\theta_{yr|c} = \theta_{yr|\bar{c}}$ and $\theta_{rc|y} = \theta_{rc|\bar{y}}$. Obviously, if we have any of the three conditional independence cases, the associations are homogeneous.

We may also consider the tables of doubly conditional probabilities:

$$\begin{pmatrix} p_{y|rc} & p_{y|r\bar{c}} \\ p_{\bar{y}|rc} & p_{\bar{y}|r\bar{c}} \end{pmatrix}$$

indicating the probabilities of y given the four different combinations of the two binary valued features, i.e.,

$$p_{y|rc} = \frac{n_{yrc}}{n_{yrc} + n_{\bar{y}rc}}$$

Again, due to the identity (1), i.e.:

$$p_{\bar{y}|rc} = 1 - p_{y|rc}$$

these two tables are redundant and the second table can be obtained from the first one. Alternatively, we may consider the corresponding table of doubly conditional odds

$$\begin{pmatrix} o_{y|rc} & o_{y|r\bar{c}} \\ o_{\bar{y}|rc} & o_{\bar{y}|r\bar{c}} \end{pmatrix}$$

where

$$o_{y|rc} = \frac{p_{y|rc}}{1 - p_{y|rc}}$$

or in terms of counts as:

$$o_{y|rc} = \frac{n_{yrc}}{n_{\bar{y}rc}}$$

Note that since we also have the identity:

$$p_{yc|r} = p_{y|rc}p_{c|r}$$

we have the following alternative expressions for the conditional odds-ratio (3):

$$\theta_{yc|r} = \frac{p_{yc|r}p_{\bar{y}\bar{c}|r}}{p_{\bar{y}c|r}p_{y\bar{c}|r}} = \frac{p_{y|rc}p_{\bar{y}|r\bar{c}}}{p_{\bar{y}|rc}p_{y|r\bar{c}}} = \frac{o_{y|rc}}{o_{y|r\bar{c}}}$$

and for the ratio of odds-ratios (4) in terms of the doubly conditional odds as:

$$\frac{\theta_{yc|r}}{\theta_{yc|\bar{r}}} = \frac{o_{y|rc}o_{y|\bar{r}\bar{c}}}{o_{y|r\bar{c}}o_{y|\bar{r}c}}$$

Accordingly, the conditional independence and the homogeneity of associations can also be checked from the table of doubly conditioned odds.

1.1 Interactions

We say that if the above considered association are not homogeneous i.e. (4) is not equal to 1, then there is an interaction on the odds-ratio scale.

We may also consider similar ratio of ratios on the doubly conditional probabilities, i.e.:

$$\frac{p_{y|rc}p_{y|\bar{r}\bar{c}}}{p_{y|r\bar{c}}p_{y|\bar{r}c}} \quad (5)$$

If this quantity equals 1, then there is an interaction on the ratio of proportions scale. We may use Bayes theorem and rewrite each doubly conditional probability as

$$p_{y|rc} = \frac{p_{rc|y}p_y}{p_{rc}}$$

Plugging these back to (5) gives us

$$\frac{p_{rc|y}p_{\bar{r}\bar{c}|y}}{p_{r\bar{c}|y}p_{\bar{r}c|y}} \frac{p_{r\bar{c}}p_{\bar{r}c}}{p_{rc}p_{\bar{r}\bar{c}}}$$

Accordingly, (5) being equal to 1 implies that the marginal odds ratio θ_{rc} must equal to the conditional odds-ratio $\theta_{rc|y}$ but it still does not necessarily equal to the conditional odds-ratio $\theta_{rc|\bar{y}}$.

Not sure whether there is any use for this.